

LECTURE 13: LOCAL AVERAGE TREATMENT EFFECTS

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OUTLINE

1. Potential Outcomes
2. Local Average Treatment Effects
3. Effects of Military Service on Earnings

POTENTIAL OUTCOMES

- Here we investigate the interpretation of instrumental variables estimators allowing for general heterogeneity in the effect of the endogenous regressor.
- The general theme is that with heterogeneous effects endogeneity creates severe problems for identification of population parameters.
- We can **only** identify average effects for subpopulations that are induced by the instrument to change the value of the endogenous regressors.
- We refer to such subpopulations as **compliers**.
- These complier subpopulations are not always the subpopulations that are *ex ante* the most interesting subpopulations, but there the data may just not be informative about average effects for other subpopulations without extrapolation.

POTENTIAL OUTCOMES

- Let $Y_i(C)$ and $Y_i(T)$ be two potential outcomes for unit i , one for each value of the endogenous regressor or treatment. We are interested in the causal effect of military service, $Y_i(T) - Y_i(C)$.
- We observe the treatment W_i and

$$Y_i \equiv Y_i(W_i) = \begin{cases} Y_i(T) & \text{if } W_i = T \\ Y_i(C) & \text{if } W_i = C. \end{cases}$$

- Define two potential outcomes $W_i(0)$ and $W_i(1)$, representing the value of the endogenous regressor given the two values for the instrument.
- The actual value of W_i is

$$W_i \equiv W_i(Z_i) = \begin{cases} W_i(1) & \text{if } Z_i = 1 \\ W_i(0) & \text{if } Z_i = 0. \end{cases}$$

- So in the end we observe Z_i , $W_i = W_i(Z_i)$ and $Y_i = Y_i(W_i(Z_i))$.

ASSUMPTIONS

- The key instrumental variables assumption is

Assumption (Independence)

$$Z_i \perp \left(Y_i(C), Y_i(T), W_i(C), W_i(T) \right).$$

- It requires that the instrument does not directly affect the outcome. The assumption is formulated in a nonparametric way, without definitions of residuals that are tied to functional forms.

ASSUMPTIONS

- It is important to note that this assumption is not implied by random assignment of Z_i . To see this an alternative formulation of the assumption is in two parts. First we postulate the existence of four potential outcomes, $Y_i(z, w)$.

Assumption (Random Assignment)

$$Z_i \perp (Y_i(0, 0), Y_i(0, 1), Y_i(1, 0), Y_i(1, 1), W_i(C), W_i(T)).$$

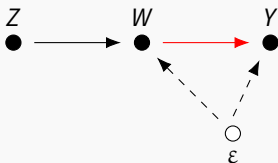
Assumption (Exclusion Restriction)

$$Y_i(z, w) = Y_i(z', w),$$

for all z, z', w .

- The first of these two assumptions is implied by random assignment of Z_i , but the second is substantial, and randomization has no bearing on it.

INSTRUMENTAL VARIABLES: DIRECTED ACYCLICAL GRAPH REPRESENTATION



- Effect of W on Y is not identified (we knew that from a previous lecture).
- No obvious way to incorporate monotonicity in the graph
- No clear way of expressing **identification** of local average treatment effect.

COMPLIANCE TYPES

- It is useful to think about the compliance behavior of the different units, that is how they respond to different values of the instrument.

COMPLIANCE TYPES		$W_i(0)$	
		C	T
$W_i(1)$	C	never-taker	defier
	T	complier	always-taker

- We cannot directly establish the type of a unit based on what we observe for them since we only see the pair (Z_i, W_i) , not the pair $(W_i(0), W_i(1))$.
- Nevertheless, we can rule out some possibilities.

COMPLIANCE TYPES

- Table 2 summarizes the information about compliance behavior from **observed** treatment status and instrument.

COMPLIANCE TYPE BY TREATMENT AND INSTRUMENT		
	Z_i	
	0	1
W_i	C complier /never-taker	never-taker/defier
	T always-taker/defier	complier /always-taker

MONOTONICITY

- To make additional progress we consider a monotonicity assumption:

Assumption (Monotonicity/No-Defiers)

$$W_i(1) \geq W_i(0).$$

This assumption makes sense in a lot of applications, and is implied directly by many latent index models of the type:

$$W_i(z) = \mathbf{1}_{\pi_0 + \pi_1 \cdot z + \varepsilon_i > 0}.$$

- For example, if Z_i is assignment to a treatment, and W_i is an indicator for receipt of treatment, it makes sense that there are few if any individuals who always do the exact opposite of what their assignment is.

MONOTONICITY

- When is monotonicity plausible and when is it not?
 - **plausible**: if instrument is an “incentive” (like draft eligibility, assignment to treatment).
 - **not plausible**: if instrument is assignment to administrator (like judge design)

COMPLIANCE TYPES GIVEN MONOTONICITY

- Given this monotonicity assumption the information we can extract from observed compliance behavior **increases**.

COMPLIANCE TYPE BY TREATMENT AND INSTRUMENT GIVEN MONOTONICITY

		Z_i	
		0	1
W_i	C	complier/never-taker	never-taker
	T	always-taker	complier/always-taker

IDENTIFYING THE COMPLIANCE TYPE DISTRIBUTION

- Let π_c , π_n , and π_a be the population proportions of compliers, never-takers and always-takers respectively.
- We can estimate those from the population distribution of treatment and instrument status:

$$\mathbb{E}[W_i|Z_i = 0] = \pi_a,$$

$$\mathbb{E}[W_i|Z_i = 1] = \pi_a + \pi_c.$$

WHAT CAN WE LEARN ABOUT OUTCOMES BY COMPLIANCE TYPE?

- Now consider average outcomes by instrument and treatment status:

$$\begin{aligned}\mathbb{E}[Y_i|W_i = C, Z_i = 0] &= \frac{\pi_c}{\pi_c + \pi_n} \cdot \mathbb{E}[Y_i(C)|\text{complier}] \\ &\quad + \frac{\pi_n}{\pi_c + \pi_n} \cdot \mathbb{E}[Y_i(C)|\text{never - taker}], \\ \mathbb{E}[Y_i|W_i = C, Z_i = 1] &= \mathbb{E}[Y_i(C)|\text{never - taker}], \\ \mathbb{E}[Y_i|W_i = T, Z_i = 0] &= \mathbb{E}[Y_i(T)|\text{always - taker}], \\ \mathbb{E}[Y_i|W_i = T, Z_i = 1] &= \frac{\pi_c}{\pi_c + \pi_a} \cdot \mathbb{E}[Y_i(T)|\text{complier}] \\ &\quad + \frac{\pi_a}{\pi_c + \pi_a} \cdot \mathbb{E}[Y_i(T)|\text{always - taker}].\end{aligned}$$

WHAT CAN WE LEARN ABOUT OUTCOMES BY COMPLIANCE TYPE?

- From these relationships we can figure out the average outcome by treatment status for compliers, and thus the average effect for compliers, or the local average treatment effect:

$$\text{LATE} = \mathbb{E}[Y_i(T) - Y_i(C)|\text{complier}]$$

- We can also get there another way. Consider the least squares regression of Y on Z_i . The slope coefficient is

$$\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]$$

- Consider the first term:

$$\begin{aligned}
 \mathbb{E}[Y_i|Z_i = 1] &= \mathbb{E}[Y_i|Z_i = 1, \text{complier}] \cdot \Pr(\text{complier}|Z_i = 1) \\
 &\quad + \mathbb{E}[Y_i|Z_i = 1, \text{never - taker}] \cdot \Pr(\text{never - taker}|Z_i = 1) \\
 &\quad + \mathbb{E}[Y_i|Z_i = 1, \text{always - taker}] \cdot \Pr(\text{always - taker}|Z_i = 1) \\
 &= \mathbb{E}[Y_i(T)|\text{complier}] \cdot \pi_c \\
 &\quad + \mathbb{E}[Y_i(C)|\text{never - taker}] \cdot \pi_n + \mathbb{E}[Y_i(T)|\text{always - taker}] \cdot \pi_a.
 \end{aligned}$$

WHAT CAN WE LEARN ABOUT OUTCOMES BY COMPLIANCE TYPE?

- Similarly

$$\mathbb{E}[Y_i|Z_i = 0] = \mathbb{E}[Y_i(C)|\text{complier}] \cdot \pi_c$$

$$+ \mathbb{E}[Y_i(C)|\text{never-taker}] \cdot \pi_n + \mathbb{E}[Y_i(T)|\text{always-taker}] \cdot \pi_a.$$

Hence the difference is

$$\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0] = \mathbb{E}[Y_i(T) - Y_i(C)|\text{complier}] \cdot \pi_c.$$

THE LOCAL AVERAGE TREATMENT EFFECT

- The same argument can be used to show that the slope coefficient in the regression of W_i on Z_i is

$$\mathbb{E}[W_i|Z_i = 1] - \mathbb{E}[W_i|Z_i = 0] = \pi_c.$$

- Hence the instrumental variables estimand is the ratio, equal to the local average treatment effect $\mathbb{E}[Y_i(T) - Y_i(C)|\text{complier}]$.
- The key insight is that the data are informative about the average effect for compliers only:
 - the data are not informative about the average effect for nevertakers because they are never seen receiving the treatment,
 - the data are not informative about the average effect for alwaystakers because they are never seen without the treatment.

BOUNDS ON THE AVERAGE TREATMENT EFFECT (MANSKI (1990))

- Writing the average effect as

$$\begin{aligned} E[Y_i(1) - Y_i(0)] &= E[Y_i(1) - Y_i(0)|C_i = a]\text{pr}(C_i = a) + E[Y_i(1) - Y_i(0)|C_i = n]\text{pr}(C_i = n) \\ &\quad + E[Y_i(1) - Y_i(0)|C_i = c]\text{pr}(C_i = c) \end{aligned}$$

- we can estimate all components other than

$$E[Y_i(0)|C_i = a] \quad \text{and} \quad E[Y_i(1)|C_i = n]$$

- Hence the lower bound on $E[Y_i(1) - Y_i(0)]$ is

$$E[Y_i(1) - 1|C_i = a]\text{pr}(C_i = a) + E[Y_i(1) - Y_i(0)|C_i = n]\text{pr}(C_i = n) + E[0 - Y_i(0)|C_i = c]\text{pr}(C_i = c)$$

and the upper bound on $E[Y_i(1) - Y_i(0)]$ is

$$E[Y_i(1) - 0|C_i = a]\text{pr}(C_i = a) + E[Y_i(1) - Y_i(0)|C_i = n]\text{pr}(C_i = n) + E[1 - Y_i(0)|C_i = c]\text{pr}(C_i = c)$$

WHAT DOES AN UNCONFOUNDEDNESS ANALYSIS DO?

- Suppose we use Z_i as a **control variable** instead of an **instrument**:

$$\mathbb{E}[Y_i|W_i = T, Z_i = 1] - \mathbb{E}[Y_i|W_i = C, Z_i = 1]$$

$$= \mathbb{E}[Y_i(T)|W_i = T, Z_i = 1] - \mathbb{E}[Y_i(C)|W_i = C, Z_i = 1]$$

$$= \left(\frac{\pi_a}{\pi_a + \pi_c} \mathbb{E}[Y_i(T)|a] + \frac{\pi_c}{\pi_a + \pi_c} \mathbb{E}[Y_i(T)|c] \right) - \mathbb{E}[Y_i(C)|n]$$

- This is **not** a valid causal comparison.

EFFECTS OF MILITARY SERVICE ON EARNINGS

- Angrist (1989) was interested in estimating the effect of serving in the military on earnings.
- There is obvious concern about the fact that those choosing to serve in the military are likely to be different from those who don't.
- Which way do you expect the bias to go?
- To avoid such biases he exploited the Vietnam era draft lottery. Specifically he uses the binary indicator whether or not your draft lottery number made you eligible to be drafted as an instrument.
- This was tied to day of birth, so more or less random. Even so, that does **not** necessarily make it valid as an instrument.
- What are potential reasons for this instrument not to be valid?
- As the outcome of interest we take log earnings.

- The simple ols regression leads to:

$$\widehat{\log(\text{earnings})}_i = 5.4364 - 0.0205 \cdot \widehat{\text{veteran}}_i$$

(0079) (0.0167)

TREATMENT STATUS BY ASSIGNMENT

		Z_i	
		0	1
W_i	C	5948	1915
	T	1372	865

COMPLIANCE TYPE DISTRIBUTION

- Using these data we get the following proportions of the various compliance types, given in the table.
- For example, the proportion of nevertakers is estimated as the conditional probability of $W = 0$ given $Z = 1$:

$$\Pr(\text{nevertaker}) = \frac{1915}{1915 + 865}.$$

COMPLIANCE TYPES: ESTIMATED PROPORTIONS		
		$W_i(0)$
		C T
$W_i(1)$	C	never-taker (0.6888) defier (C)
	T	complier (0.1237) always-taker (0.1875)

AVERAGE OUTCOMES BY TREATMENT TYPE AND INSTRUMENT

- The table below gives the average outcomes for the four groups, by treatment and instrument status.

ESTIMATED AVERAGE OUTCOMES BY TREATMENT AND INSTRUMENT

		Z_i	
		0	1
W_i	C	$\widehat{\mathbb{E}[Y_i]} = 5.4472$	$\widehat{\mathbb{E}[Y_i]} = 5.4028$
	T	$\widehat{\mathbb{E}[Y_i]} = 5.4076,$	$\widehat{\mathbb{E}[Y_i]} = 5.4289$

AVERAGE OUTCOMES BY COMPLIANCE TYPE

- The table below gives the estimated averages for the four **compliance** types.

COMPLIANCE TYPES: ESTIMATED AVERAGE OUTCOMES

		$W_i(C)$	
		C	T
$W_i(T)$	C	never-taker: $\widehat{E}[Y_i(C)] = 5.4028$	defier (N
	T	complier: $\widehat{E}[Y_i(C)] = 5.6948, \widehat{E}[Y_i(T)] = 5.4612$	always-taker: $\widehat{E}[Y_i(C)] = 5.4612, \widehat{E}[Y_i(T)] = 5.6948$

- The **local average treatment effect** is -0.2336, a 23% drop in earnings as a result of serving in the military.

COMPARISON WITH TSLS

- Simply doing IV or TSLS would give you the same number:

$$\begin{array}{rcccl} \log(\widehat{\text{earnings}})_i & = & 5.4836 & - & 0.2336 \cdot \widehat{\text{veteran}}_i \\ & & (0.0289) & & (0.1266) \end{array}$$

MULTIVALUED/CONTINUOUS TREATMENTS

- Example: years of education in Angrist-Krueger study.
 - outcome: log earnings
 - treatment years of education
 - instrument: indicator for quarter of birth
- Monotonicity: $S_i(1) \geq S_i(0)$ (changing quarter can only increase schooling, not lower it)
- testable assumption

MULTIVALUED/CONTINUOUS TREATMENTS

- result, assuming endogenous regressor takes on $J + 1$ values, $S_i \in \{0, 1, \dots, J\}$, the IV estimand is

$$\frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[S_i|Z_i = 1] - \mathbb{E}[S_i|Z_i = 0]} = \sum_{j=1}^J \omega_j \mathbb{E}[Y_i(j) - Y_i(j-1) | S_i(1) \geq j > S_i(0)]$$

- with weights ω_j

$$\omega_j = \frac{\text{pr}(S_i(1) \geq j > S_i(0))}{\sum_{k=1}^J \text{pr}(S_i(1) \geq k > S_i(0))}$$

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